

3R4L — the software stack and the setup from scratch (a Mac)

Written 2026-10-10 by the build seat from the site's own files, this Mac's toolchain and the build records (five read-only readers → one writer → a clarity critic → the final). What we used, what a friend would use, and the commands in order. The site: <https://stevy888.github.io/3r4l/> · the repo: github.com/stevy888/3r4l.

1. What we used

The site in one line: 3r4l.org is a static two-language card site, about 8 pages, no backend, each page ≈ 36–59 KB with its CSS and JS inlined, served as plain files from a `docs/` folder on GitHub Pages. One Node script builds it; nothing else runs in production.

Layer	What	Version	Why
OS	macOS on Apple silicon	26.2 (Darwin 25.2.0)	Any Apple-silicon Mac runs this; the M3 Ultra here is not needed
Shell / terminal	zsh in Apple Terminal.app	zsh 5.9	Stock; no iTerm, VS Code or Cursor installed
Package manager	Homebrew	7.0.4	Every CLI below comes from it; needs Xcode Command Line Tools
Editor / harness	Claude Code CLI	2.1.296	The editor and the pair of hands; nothing else edits the repo
Version control	git · GitHub CLI <code>gh</code>	2.53.0 · 2.89.0	<code>gh</code> login is the one deploy credential
Runtime	Node	25.9.0 (needs ≥ 20)	Runs the build, the wording check, the screenshot tests, the mail scripts
Site build	<code>build.mjs</code> , one ESM script, zero npm deps	—	Reads <code>content/</code> , writes <code>docs/</code> for en + tl; inlines CSS/JS; writes <code>sw.js</code> , <code>manifest.webmanifest</code> , <code>robots.txt</code> , <code>.nojekyll</code>
CSS / JS	Hand-written <code>content/page.css</code> , <code>day.js</code> , <code>theme.js</code>	—	No framework, bundler or CSS tool; comments stripped at build to stay under the byte limits
Fonts	Source Serif 4, self-hosted WOFF2 400 / 400i / 500 (OFL 1.1)	font v4.327	Three subsetted files ≈ 15 KB each; no Google Fonts request
Phone-screenshot tests	<code>@playwright/test</code> + bundled Chromium	1.59.1 (Chromium rev 1217)	<code>measure.mjs</code> , <code>offline-proof.mjs</code> , <code>print-proof.mjs</code> , <code>frames-plate.mjs</code>

Layer	What	Version	Why
Audio-decode test	Google Chrome (optional)	155	<code>prove-reading.mjs</code> prefers Chrome for a real AAC decoder; without it the audio-decode row is skipped
Wording check	<code>check-wording</code> (bash + node, zero deps)	—	Every text node must be in the page's allowlist; no <code><script src></code> , <code></code> , iframes, trackers
PDF text	poppler <code>pdftotext</code>	26.07.0	Extracts the card's words once; never retyped
Print PDFs	Python venv: PyMuPDF (fitz) · segno · Pillow	3.14.6 · 1.28.2 · 1.6.6 · 12.3.0	<code>kit/make_kit.py</code> imposes the card N-up with cut marks; QR code
HTML → PDF	Playwright Chromium <code>page.pdf</code>	1.59.1	<code>kit/make-forty-sheet.mjs</code> writes <code>docs/forty.pdf</code>
Images	<code>cwebp</code> (webp) · macOS <code>sips</code> · Pillow	webp 1.6.0	Card faces as lossless WebP; ImageMagick not installed
Media	ffmpeg · whisper.cpp · ElevenLabs TTS	8.1.2 · — · <code>eleven_multilingual_v2</code>	One narration file <code>docs/film/card.m4a</code> (AAC, 2:27); whisper checks the spoken words against <code>card.txt</code>
Hosting	GitHub Pages from <code>docs/</code> on <code>main</code> , public repo	—	Deploy = <code>git push</code> ; no CI, no server
Domain / DNS / inbound mail	Cloudflare Registrar + DNS + Email Routing (planned)	—	3r4l.org not yet registered; hello@ / privacy@ forwarding is free
Mail list	Kit (ex-ConvertKit) free plan, API v4	—	One 41-mail sequence loaded by <code>mail/load-sequence.mjs</code> ; key awaited
AI, interactive	Claude Code on Fable 5.1 at effort xhigh	<code>claude-fable-5-1</code>	Design, build and judgement sessions
AI, blind scoring	Opus 5.5 on the Message Batches API	<code>claude-opus-5-5</code>	10 lenses × 3 reads per screenshot set; \$1–2.3 a round
Secrets	<code>~/.secrets/<project>.env</code> , gitignored, read in a subshell	—	Never in the repo

Twelve words, translated once

- *wording check* — the script that refuses any page text not in the files of record
- *phone-screenshot test* — Playwright opens every page at 320×568 / 390×844, light and dark, and asserts the limits
- *byte limits* — every page ≤ 60 KB; home + any one other ≤ 120 KB
- *fold rules* — what must be visible on the first screen without scrolling
- *doors* — links and buttons (≥ 44 px tall)

- *texts of record* — `card.txt`, `sheet.json`, `sheet2.json`, `verses.json`: the only source of any word on the site
- *the owner's eye* — you look at it and say yes
- *living card* — the card page that reads itself aloud and lights each line as it is spoken
- *grey-cloud* — a Cloudflare DNS row with the proxy OFF (DNS only)
- *N-up* — several cards per sheet, with cut marks
- *WER* — word error rate of the spoken audio against the written card
- *k=3 / lenses / sealed shuffle* — each screenshot judged 3 times on 10 questions, with labels shuffled so the judge cannot tell which is which

2. The friend's stack from scratch (Mac)

Accounts to open, in order

1. GitHub — free (Pages on a public repo is free; a *private* repo needs GitHub Pro, paid)
2. Claude subscription for Claude Code — tier and monthly price: not on record here; check claude.ai/pricing
3. Anthropic Console API key + prepaid credit — only for blind AI scoring; minimum top-up not on record
4. Cloudflare — free account; the .org at registrar cost ($\approx$ \$10–12/yr for one)
5. Kit (kit.com) — free plan (10,000 subscribers, double opt-in on by default)
6. ElevenLabs — only to re-record the audio; price not on record. macOS `say` is the free stand-in
7. Suno via kie.ai — optional music bed, $\approx$ \$0.66 a track

Layer	Recommended	Version	Cost	Why
Machine	Any Apple-silicon Mac	macOS 14+	—	The build is tiny; 8 GB RAM is plenty
Shell / terminal	zsh in Terminal.app	stock	free	Same as ours; nothing to install
Package manager	Homebrew	current	free	One command per tool
Editor / harness	Claude Code CLI	current (we ran 2.1.296)	subscription	Reads, edits, runs the scripts; you review
Version control	git + <code>gh</code>	current	free	<code>gh</code> logs in once and is the deploy credential
Runtime	Node, current Homebrew formula (<code>brew install node</code>)	$\geq$ 20 (we ran 25.9)	free	The scripts only need <code>node:fs/path/crypto</code> + global <code>fetch</code> ; current is fine
Site build	Clone our <code>build.mjs</code> + <code>content/</code>	—	free	Simpler than any framework; one file, no deps. (Eleventy/Astro would work but add nothing here)

Layer	Recommended	Version	Cost	Why
Fonts	Source Serif 4 WOFF2 400/400i/500 — already in the clone	4.327	free (OFL)	To re-make: google-webfonts-helper → Source Serif 4, latin, 400/400i/500, or @fontsource/source-serif-4; ship the OFL licence beside them
Phone tests	@playwright/test + Chromium	pin @playwright/test@1.59	free	What the tests were proven on
Audio test	Google Chrome	any current	free	Optional; without it the audio-decode row is skipped
Wording check	check-wording from the clone	—	free	Most friends would skip it; keep it if you print the card — it stops a retyped word
PDF text	poppler	current	free	pdftotext once per card PDF
Print PDFs	Python 3.14 venv: pymupdf, segno, pillow	3.14	free	Only if you print cards; skip otherwise
Images	webp (cwebp) + sips	current	free	No ImageMagick needed
Media	ffmpeg; say or ElevenLabs; whisper-cpp optional	current	free / paid TTS	The clone already holds the finished card.m4a; re-recording is optional
Hosting	GitHub Pages from docs/ (public repo)	—	\$0	Push = deploy. Cloudflare Pages only if the repo must be private and you will not pay for Pro
Domain + DNS + inbound mail	Cloudflare Registrar, DNS (grey-cloud), Email Routing	—	≈ \$10–12/yr	At-cost registrar, free forwarding for hello@/privacy@
Mail list	Kit free, plain <form method=post>	API v4	free	One sequence, 41 mails, loadable by script; or paste them in Kit's UI
AI, interactive	Claude Code, /model claude-fable-5-1, effort xhigh	—	subscription	Our design and build sessions
AI, blind scoring	claude-opus-5-5 on the Message Batches API (\$2 in / \$10 out per Mtok)	SDK ≥ 0.79	\$1–2.3 a round	Optional; your own eye is free
Secrets	~/.secrets/3r4l.env, gitignored, read in a subshell	—	free	(set -a; ~/.secrets/3r4l.env; set +a; node mail/load-sequence.mjs --dry-run)

Two edits the clone needs today (not yet made in our repo):

- `build.mjs` line 14 reads `/Users/aibrain/kj-rebuild/.../legacy-www-redirects.json` — on a fresh clone `node build.mjs` dies with `ENOENT`. Copy that JSON to `content/redirects.json` and read it from there (or guard with `existsSync` → `{rows:[]}`).
- `build.mjs` line 10 and `check-wording` line 20 hard-code `/Users/aibrain/projects/3r4l-inputs`. Make it `process.env.INPUTS || './3r4l-inputs'`. The folder may be empty; it may hold `card-of-record.pdf`, `dns-rows.txt`, `pastor-check.txt`, `card-tl.txt`, `controller.txt` — each only unlocks an extra feature.

3. Setup on a fresh Mac, in order

Budget about 30–60 minutes online; the Chromium download is the largest piece (a few hundred MB; not measured here). Lines marked *optional* can wait.

```

xcode-select --install                                # Command Line Tools (Homebrew
needs them)
/bin/bash -c "$(curl -fsSL
https://raw.githubusercontent.com/Homebrew/install/HEAD/install.sh)" #
Homebrew
brew install git gh node python@3.14 poppler ffmpeg webp # node ≥ 20 is enough; poppler =
pdftotext; webp = cwebp
brew install --cask google-chrome                    # optional: real AAC decoder
for prove-reading
brew install whisper-cpp                             # optional: only to re-check re-
recorded audio
gh auth login                                          # GitHub; this is the deploy
credential
npm i -g @anthropic-ai/claude-code && claude          # Claude Code; then /login with
your subscription
# in Claude Code: /model claude-fable-5-1 - effort is set in ~/.claude/settings.json
("effortLevel": "xhigh")
gh repo fork stevy888/3r4l --clone && cd 3r4l        # clone, don't git init
(content/, fonts, card.m4a all come with it)
# make the two path edits from $2 (redirects.json; INPUTS env var) before the first
build
npm init -y && npm i -D @playwright/test@1.59         # creates package.json +
package-lock.json; commit both (node_modules is gitignored)
npx playwright install chromium                      # Playwright's own Chromium
build (ours is a symlinked node_modules; yours is real)
python3.14 -m venv .venv && .venv/bin/pip install pymupdf segno pillow anthropic #
python3.14 explicitly: Apple's python3 has no pymupdf wheel; anthropic only for
blind scoring
./doctor                                              # expect OK on: node,
@playwright/test+chromium, venv, pdftotext, gh auth (+ getbible if online). The
token-file and two staging lines FAIL for you - fine, they serve only '--
refresh`
mkdir -p ../serve && ln -s "$PWD/docs" ../serve/3r4l # measure and probe-links
expect the site under /3r4l/, exactly as github.io/<repo>/ serves it
node build.mjs                                       # content/ → docs/
bash check-wording docs/                           # wording check; must end rc 0
# --- second Terminal tab (Ctrl-C stops it) ---
cd ../serve && python3 -m http.server 8168 --bind 127.0.0.1
# --- back in the first tab ---
node measure.mjs http://127.0.0.1:8168/3r4l/        # phone-screenshot test; rc 1 on
any red
node prove-reading.mjs http://127.0.0.1:8168/3r4l/  # audio test (prints 'decoder
none' without Chrome; the decode row is skipped)
node offline-proof.mjs http://127.0.0.1:8168/3r4l/  # offline worker
bash probe-links http://127.0.0.1:8168/3r4l/        # every internal link answers
200 (see the script's header for its argument)
git add -A && git commit -m "first build" && git push # deploy
gh api -X POST repos/<you>/3r4l/pages -f 'source[branch]=main' -f 'source[path]=/docs'
# first Pages deploy; live at https://<you>.github.io/3r4l/ in minutes
git revert HEAD && git push                          # rollback of a bad deploy;
Pages re-serves in minutes

```

Later, when you want them

```

# Custom domain (after buying the .org at Cloudflare Registrar, auto-renew ON)
# Cloudflare DNS, every row grey-cloud: A 185.199.108.153 / .109.153 / .110.153 /
  .111.153 ; CNAME www → <you>.github.io ; TXT _github-pages-challenge-<you>
  (GitHub shows the value)
gh api -X PUT repos/<you>/3r4l/pages -f cname=3r4l.org # GitHub then commits
  docs/CNAME — pull it; build.mjs writes none, so keep it in docs/
gh api -X PUT repos/<you>/3r4l/pages -F https_enforced=true # only after the
  certificate issues (up to 24 h); github.io is HTTPS already
# Email Routing (hello@, privacy@, dmarc@) needs the zone on Cloudflare DNS; add SPF,
  the vendor's DKIM CNAMEs and _dmarc as the vendor states

# Mail list (Kit)
# Kit: create a form, turn double opt-in on, copy its form_id and action URL into
  content/signup.json as {action, method:"post", fields:{email:"email_address",
  lang|none, honeypot|none}, lands_on, form_id, vendor:"kit"}
echo 'KIT_API_KEY=...' > ~/.secrets/3r4l-kit.env && chmod 600 ~/.secrets/3r4l-kit.env
node mail/build-mails.mjs && bash mail/check-mails # 41 plain-text mails, fenced
(set -a; . ~/.secrets/3r4l-kit.env; set +a; node mail/load-sequence.mjs --dry-run) #
  dry run first; a live load also needs KIT_LIVE=yes

# Print kit (needs INPUTS/card-of-record.pdf; skipped without it; output stays private
  until --publish)
.venv/bin/python kit/make_kit.py
node kit/make-forty-sheet.mjs --lang en --out docs/forty.pdf

# Audio: skip — docs/film/card.m4a is in the clone. film/run-chain.sh depends on scripts
  outside this repo (the KJ studio) and an ElevenLabs key; to re-record use `say`
  + ffmpeg or your own TTS

```

4. How a page is built and proven

Every change goes through the same six steps, cheapest first; nothing is pushed that a script has not passed and you have not seen. The blind-scoring script (`panel.py`) is **not in the repo** — it lives in a private working folder — so a friend either looks with their own eye or writes ~100 lines against the Messages API (`pip install anthropic`, `ANTHROPIC_API_KEY` in `~/.secrets`, model `claude-opus-5-5`, send each screenshot with the ten questions, three reads each, labels shuffled).

1. **Build** — `node build.mjs`; `content/` → `docs/`; the byte limits fail the build if a page is over 60 KB.
2. **Wording check** — `bash check-wording docs/`; every text node is in the files of record, no scripts, images or trackers; rc 0 or stop.
3. **Phone-screenshot test** — serve `../serve` on :8168, `node measure.mjs <base> --frames frames/`; 36 rows at four sizes, light/dark, three passes; `markers/measures.json` must say red 0.
4. **The other proofs** — `prove-reading.mjs` (audio cues), `offline-proof.mjs`, `print-proof.mjs`, `bash probe-links`; each rc 0.
5. **Blind read (optional)** — Opus 5.5 on the Batch API, ≈ \$1–2.3 a round; the bar we chased was 9.5/10 (best so far 8.64). Not shipped — use your own eye if you skip it.
6. **Your eye, then push** — open the local pages on your phone; if yes, `git add -A && git commit && git push`; Pages serves in minutes; `git revert HEAD && git push` undoes it.

5. What to skip, and what it costs

Skip

- Frameworks and bundlers (React, Next, Astro, Vite): one build script and inlined CSS/JS keep every page under 60 KB.
- A CMS: the texts of record are six small files in `content/`; Claude Code edits them.
- Analytics and trackers: by design none; the wording check refuses `gtag`, `fbq()`, `fetch()`, cookies and any external request.
- A form builder: sign-up is one plain `<form method=post>` to Kit; contact is links only (Messenger · Viber · Text · Email).
- A web-font service: three self-hosted WOFF2 files; no Google Fonts call.
- A canvas tool (Figma etc.): design was drawn in CSS and judged on real phone screenshots.
- Our estate machinery (spend fences, shared-tree landing scripts, sub-agent fleets, the KJ studio): governance for many parallel sessions; a friend uses `git commit && git push` and one Claude Code session.

Running costs | Item | Cost | ---|---| | Hosting (GitHub Pages, public repo) | \$0 | | Domain, one .org at Cloudflare Registrar | ≈ \$10–12/yr (three redirecting domains ≈ \$34/yr; Porkbun \$7.98 first year) | | Inbound mail (Cloudflare Email Routing) | \$0 | | Mail list (Kit free, ≤ 10,000 subscribers) | \$0 (Kit badge stays on the mails) | | Claude subscription for Claude Code | monthly; tier and price not on record here | | AI blind read, one round (Opus 5.5 Batch, 120–150 reads) | ≈ \$1–2.3 | | AI design round with three AI-drawn rivals + two judges | ≈ \$50–75 (we stopped doing this; one session drawing by hand + a \$1–3 blind read was as good) | | A full build day in Claude Code with sub-agents | ≈ \$67–100 measured — a friend on a subscription needs only the subscription plus \$1–3 per blind round | | Audio re-record (ElevenLabs) | price not on record; `say` is free | | Music bed (Suno via kie.ai), optional | ≈ \$0.66 a track |

Critic's verdict: FIX.